Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To study different word embedding for representation of textual data in vectorized form	
Experiment No: 8	**Date:**	**Enrolment No:92301733021**

Aim: To study different word embedding for representation of textual data in vectorized form

IDE: Google Colab

Theory:

Word Embedding is an approach for representing words and documents. Word Embedding or Word Vector is a numeric vector input that represents a word in a lower-dimensional space. It allows words with similar meanings to have a similar representation.

Word Embeddings are a method of extracting features out of text so that we can input those features into a machine learning model to work with text data. They try to preserve syntactical and semantic information. The methods such as Bag of Words (BOW), CountVectorizer and TFIDF rely on the word count in a sentence but do not save any syntactical or semantic information. In these algorithms, the size of the vector is the number of elements in the vocabulary. We can get a sparse matrix if most of the elements are zero. Large input vectors will mean a huge number of weights which will result in high computation required for training. Word Embeddings give a solution to these problems.

Need for Word Embedding?

- To reduce dimensionality
- To use a word to predict the words around it.
- Inter-word semantics must be captured.

How are Word Embeddings used?

- They are used as input to machine learning models. Take the words —-> Give their numeric representation —-> Use in training or inference.
- To represent or visualize any underlying patterns of usage in the corpus that was used to train them.

1. Bag of Words

Bag of Words (BOW) is an algorithm that counts how many times a word appears in a document. Those word counts allow us to compare documents and gauge their similarities for applications like search, document classification, and topic modeling.

2. Tf-Idf

Tf-Idf is shorthand for term frequency-inverse document frequency. So, two things: term frequency and inverse document frequency. Term frequency (TF) is basically the output of the BoW model. For a specific document, it determines how important a word is by looking at how frequently it appears in the document. Term frequency measures the importance of the word. If a word appears a lot of times, then the word must be important. For example, if our document is “I am a cat lover. I have a cat named Steve. I feed a cat outside my room regularly,” we see that the words with the highest frequency are I, a, and cat. This agrees with our intuition that high term frequency = higher importance.

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TF(t) = (Number of times term t appears in a document) / (Total number of terms in the document)

IDF used to calculate the weight of rare words across all documents. The words that occur rarely in the corpus have a high IDF score. However, it is known that certain terms, such as “I”, “a” may appear a lot of times but have little importance. Thus we need to weigh down the frequent terms while scaling up the rare ones

IDF(t) = log_e(Total number of documents / Number of documents with term t in it)

Pre Lab Exercise:

1. **Perform BoW vectorization of the corpus**
2. **Perform TF vectorization of the corpus**

Code:

```
sentences = ["India is a beutifull country",
"India is a Powerfull country in the world ",
"Inida is a developing Country",

"Cricket my is favourite game ",
"Sport give us a chance to win the world",

"Technology is the application of scientific knowledge to solve practical problems and improve daily life",
"Technology includes tools, machines, and systems that convert natural resources into usable products"]

#IMPORTING BASIC LIBRARIES
import numpy as np
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer #TF-BoW
from sklearn.feature_extraction.text import TfidfVectorizer #TF-IDF
from sklearn.metrics.pairwise import cosine_similarity
#TF-BoW Embedding
tf_vectorizer=CountVectorizer()
tf_emb=tf_vectorizer.fit_transform(sentences)

#similarity matrix using cosine_similarity
tf_sim=cosine_similarity(tf_emb)

tf_sim
#TF-IDF Embedding
tfidf_vectorizer=TfidfVectorizer()
tfidf_emb=tfidf_vectorizer.fit_transform(sentences)
tfidf_sim=cosine_similarity(tfidf_emb)
tf_sim
# Recommend the most matching/similar sentence
def doc_sim(tfidf_sim,sentences):
```

```
for i in range(len(sentences)):
    sim = tfidf_sim[i].copy()
    sim[i]=-1
    top_index=np.argmax(sim)
    print(sentences[i],"->",sentences[top_index])

    doc_sim(tfidf_sim,sentences)
```

Output:

```
India is a beutifull country -> India is a Powerfull country in the world
India is a Powerfull country in the world  -> India is a beutifull country
Inida is a developing Country -> India is a beutifull country
Cricket my is favourite game  -> India is a beutifull country
Sport give us a chance to win the world -> India is a Powerfull country in the world
Technology is the application of scientific knowledge to solve practical problems and improve daily life -> Sport give us a chance to win the world
Technology includes tools, machines, and systems that convert natural resources into usable products -> Technology is the application of scientific knowledge to solve practical problems and improve daily life
```

Observation:

In this experiment, both BoW and TF-IDF vectorization techniques are used to convert text into numerical form and measure similarity between sentences using cosine similarity. It is observed that TF-IDF provides more meaningful similarity results compared to BoW, as it reduces the impact of common words and highlights important terms. The model successfully identifies the most similar sentence for each input, but results may be affected by spelling errors and small dataset size.

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Post Lab Exercise:

Take three documents and find similarity using

a. BoW vectorization
b. TF Vectorization
c. TF-IDF Vectorization

Comment over the answer obtained in each of the case.

a. BoW Vectorization (Observation)

Using Bag of Words, similarity is calculated based on the count of common words between documents. It is observed that documents with more shared words show higher similarity, but BoW does not consider word importance or context, which can lead to less accurate results.

b. TF Vectorization (Observation)

In Term Frequency, similarity is calculated using normalized word frequency. It improves over BoW by considering how often a word appears in a document, but still treats all words equally without considering their importance across documents.

c. TF-IDF Vectorization (Observation)

TF-IDF provides better similarity results as it gives higher importance to unique words and reduces the effect of common words. It is observed that TF-IDF captures meaningful relationships between documents more accurately compared to BoW and TF.